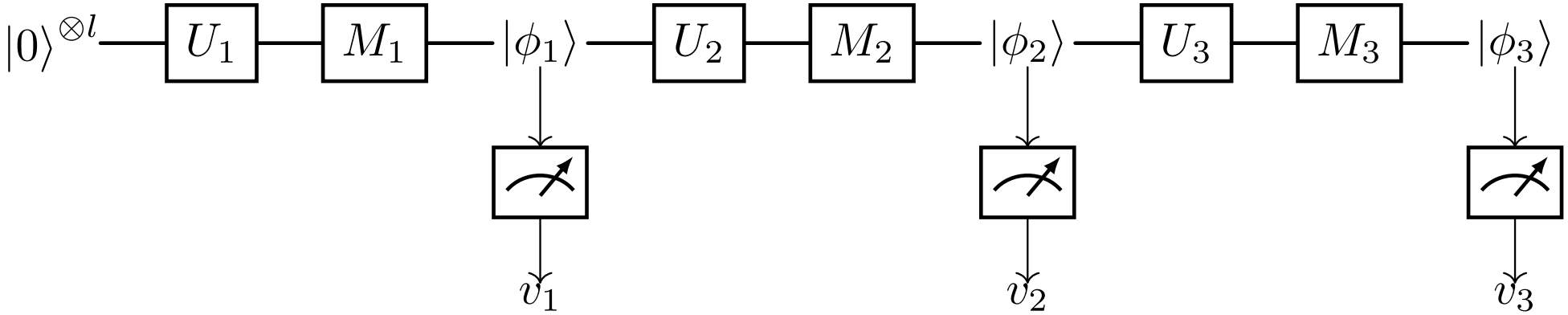
$|0\rangle^{\otimes l}$
U_1
M_1
$|\phi_1\rangle$
U_2
M_2
$|\phi_2\rangle$
U_3
M_3
$|\phi_3\rangle$
v_1
v_2
v_3